Notes: Corhay, Kung, and Schmid (JFE, 2025)
Q: Risk, rents, or growth?

Contents

1 Big picture

- **Question.** Why has Tobin's Q become increasingly polarized inside industries, with superstar firms trading at much higher valuation ratios than the rest of the firms?
- **Main empirical fact.** Superstar firms and small firms diverge strongly in Q, markups, profitability, and intangible capital shares from 1978 to 2019, but not in sales growth or discount rates.
- **Mechanism.** Superstars are oligopolists with strategic pricing power and productive intangible capital. Small firms are a monopolistically competitive fringe that invest in speculative innovation to become superstars.
- **Key result.** Rising barriers to entry at both stages explain most of the increase in markup polarization, but Q polarization is mainly driven by barriers to creating small firms, rising demand for superstar goods, and the growing role of superstar intangible capital.
- **Policy result.** Lowering barriers to becoming a superstar reduces markups and stimulates growth. Lowering barriers to creating small firms reduces Q and markup inequality but can depress speculative innovation and trend growth.

2 Data and measurement

2.1 Superstars and small firms

Superstars are firms in the top 5% of market capitalization within each 4-digit NAICS industry-year. Small firms are all remaining firms in the same industry. Polarization is the difference in an average characteristic between superstars and small firms within industries.

Interpretation. The within-industry definition compares dominant firms to their industry peers rather than comparing different sectors.

2.2 Firm-level variables

Firm-level accounting variables come from Compustat. Tobin's Q is market value of equity plus debt divided by physical capital. Physical capital is gross PPE. Intangible capital follows Peters and Taylor and is based on R&D plus a share of SG&A.

Interpretation. Because Q is scaled by physical capital, rising productive intangible capital raises Q rather than being fully netted out.

2.3 Markups

The markup is constructed from the production approach:

$$\text{markup}_{it} = \frac{\xi_{it}^{v}}{\text{share}_{it}^{v}},$$

where ξ_{it}^{v} is the output elasticity of a variable input and share_{it}^{v} is its revenue share.

Interpretation. The markup measure links the empirical trend to the model's strategic-pricing distinction between superstars and small firms.

3 Model and empirical specifications

3.1 Demand and mixed industry structure

$$\log \bar{Y}_t = \int_0^1 \log Y_{jt}\, dj, \qquad Y_{jt} = \left[\Upsilon X_{jt}^{(\nu-1)/\nu} + x_{jt}^{(\nu-1)/\nu}\right]^{\nu/(\nu-1)}.$$

Interpretation. The parameter Υ shifts demand toward superstar goods; ν governs substitution across goods.

3.2 Speculative innovation

$$h(j_{mt}) = 1 - \exp\left[-\frac{1}{\vartheta}\frac{j_{mt}}{\bar{j}_t}\right].$$

Small firms invest in speculative R&D to raise the probability of a breakthrough into superstar status.

Interpretation. A policy or parameter change that raises the reward to becoming a superstar can increase speculative innovation even if barriers are high.

3.3 Q decomposition

$$\bar{Q} - \bar{\mathcal{Q}} = \left(\frac{\bar{Z}}{\bar{K}} - \frac{\bar{z}}{\bar{k}}\right) + \left[\frac{(1-1/\bar{\Phi})SK - FC}{\bar{R} + \bar{\Delta}_n - \bar{G}} - \frac{(1-1/\bar{\varphi})sk - fc}{\bar{r} + \delta_n - \bar{g}}\right].$$

Interpretation. Q polarization can rise because of intangibles, markups, sales to capital, fixed costs, discount rates, exit rates, or growth rates.

3.4 Simulated method of moments

$$\hat{\theta} = \arg\min_{\theta}[m_{data} - m_{model}(\theta)]' W [m_{data} - m_{model}(\theta)].$$

The model is estimated separately in 1978–1998 and 1999–2019.

Interpretation. The model is forced to match both aggregate macro moments and within-industry polarization moments.

4 Identification and empirical design

- The average small-firm markup identifies the substitution elasticity ν.
- The superstar sales share identifies the superstar taste parameter Υ.
- The number of superstars relative to small firms, superstar markups, death rates, and Q differences discipline entry-barrier and transition parameters.
- Table 1 shows that identification is joint: many moments respond to multiple parameters.
- Table 5 changes groups of parameters from period-one to period-two values to decompose the sources of Q polarization.

5 Tables and figures

5.1 Figure 1: Industry Polarization

Read: Figure 1 shows that Q, markups, profitability, and intangible capital shares diverge sharply between superstars and small firms, while sales growth and discount rates do not.

Economic message. The figure motivates the paper: Q polarization is closely connected to rents and intangible capital, not only to growth or discount-rate differences.

A. Corhay et al. *Journal of Financial Economics 165 (2025) 103990*

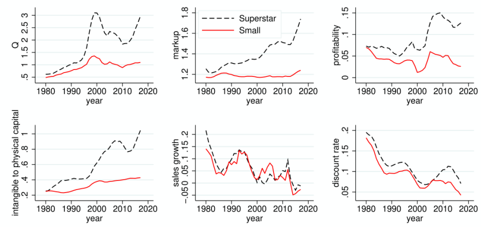

Fig. 1. Industry Polarization.
This figure plots the 5-year centered moving average of Tobin's Q, markups, profitability, intangible capital shares, sales growth, and discount rates for 1978–2019. The variable definitions are provided in internet appendix C.2. We aggregate small and superstar firms across industries and then take a rolling five-year, centered moving average. Superstars are defined as the top 5% ranked by market cap in each 4-digit NAICS industry. Small firms are defined as the remaining firms, excluding Superstars.

A. Corhay et al. *Journal of Financial Economics 165 (2025) 103990*

Table 1
Jacobian matrix.

	$a^\star$	η	α	β	γ	Θ	δ_n	κ	ϑ	ν	Υ	F	f
Output growth	0.09	0.08	0.05	1.64	−0.00	−0.02	0.00	0.00	−0.06	0.06	0.14	0.00	0.01
Intangible capital ratio	0.91	1.89	−0.80	11.23	0.06	−0.22	0.01	0.02	−0.72	0.73	1.68	−0.00	0.09
Labor share	0.10	0.01	−0.20	1.62	0.00	0.00	−0.02	−0.02	0.01	−0.02	−0.20	−0.00	−0.04
Risk-free rate	0.04	0.04	0.03	−0.52	−0.00	−0.01	−0.00	0.00	−0.03	0.03	0.07	0.00	0.00
Earnings-price ratio	−0.09	−0.07	−0.06	−4.72	0.02	0.02	0.00	−0.00	0.06	−0.03	−0.12	0.01	−0.00
Markup superstar firms	−0.21	−0.07	−0.12	−3.55	−0.00	0.04	0.04	0.03	0.10	0.16	0.40	−0.00	0.08
Firm death rate	0.00	0.00	0.00	0.01	0.00	0.00	0.02	−0.00	−0.00	−0.00	0.00	−0.00	−0.00
Relative size superstar	1.08	0.65	0.51	28.19	0.06	−0.16	−0.23	−0.20	−0.47	0.49	3.09	−0.00	−0.51
$Q-q$	2.32	0.94	1.01	54.66	0.09	−0.28	1.10	0.45	0.67	12.16	26.01	−6.24	3.83
Markup small firms	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	−0.19	0.00	0.00	0.00
Sales share superstar	−0.87	−0.18	−0.40	−8.07	0.00	−0.16	0.11	0.11	−0.55	2.00	1.76	0.00	0.35
ROA superstar firms	−0.03	−0.04	−0.02	−2.73	0.00	0.02	0.01	0.01	0.05	0.07	0.17	−0.07	0.02
ROA small firms	−0.03	−0.03	−0.02	−2.10	0.00	0.01	0.00	0.00	0.03	−0.08	−0.14	0.02	−0.02

This table shows the average of the two Jacobian matrices associated with the estimation of the model. Each entry of the matrix reports the percentage change in each moment given a one percent increase in each parameter. To avoid null values, rate moments expressed in gross terms. Details regarding the definition and construction of variables can be found in internet appendix C.

time, as the elasticities are estimated within five-year rolling windows, period with standard errors are tabulated in Panel B, and the calibration

5.2 Table 1: Jacobian matrix

Read: The table reports how target moments respond to structural parameters. Q differences are highly sensitive to demand and intangible-capital parameters, while small-firm markups respond mainly to ν.

Economic message. This is the identification map of the paper.

5.3 Table 2: Parameters description

Read: Table 2 lists estimated and calibrated parameters governing preferences, technology, entry, demand, fixed costs, depreciation, adjustment costs, and exogenous productivity.

Economic message. The mechanism is multi-dimensional: Q polarization is jointly produced by demand, innovation, entry, and asset-pricing forces.

A. Corhay et al. *Journal of Financial Economics 165 (2025) 103990*

Table 2
Parameters description.

A. Estimated parameters			
$a^\star$	Exogenous technology scale	κ	Small firm entry cost
η	Production elasticity intangible capital	ϑ	Small firm patent R&D scale
α	Production elasticity physical capital	ν	Elasticity of substitution across goods
β	Subjective discount factor	Υ	Superstar goods taste
γ	Risk aversion	F	Superstar fixed cost
Θ	Superstar strategic R&D scale	f	Small firm fixed cost
δ_n	Small firm death rate		
B. Calibrated parameters			
ψ	Intertemporal elasticity of substitution	ζ_z	Intangible capital adjustment cost
χ	Labor elasticity	ρ	Persistence of exogenous productivity
δ_k	Depreciation rate physical capital	σ	Volatility of exogenous productivity
δ_z	Depreciation rate intangible capital	Δ_0	Superstar death rate
ζ_k	Physical capital adjustment cost		

This table summarizes the definition of all model parameters.

A. Corhay et al. *Journal of Financial Economics 165 (2025) 103990*

Table 3
Target moments and parameter estimates.

Panel A: Target moments

	Data		Model	
	1978–1998	1999–2019	1978–1998	1999–2019
Mean risk-free rate	3.06%	−0.12%	3.05%	−0.13%
Mean output growth	2.10%	1.31%	2.10%	1.31%
Mean earnings-price ratio	3.56%	1.83%	3.56%	1.83%
Mean $Q-q$	0.28	1.36	0.28	1.36
Mean firm death rate	9.48%	8.27%	9.48%	8.27%
Mean markup superstar firms	28.88%	51.60%	28.89%	51.61%
Mean markup small firms	18.97%	19.07%	18.97%	19.08%
Mean intangible capital ratio	0.10	0.43	0.10	0.43
Mean sales share superstar	0.31	0.32	0.31	0.32
Mean relative size superstar	10.74	10.45	10.74	10.45
Mean labor share	0.64	0.60	0.64	0.60
Mean ROK superstar firms	6.70%	11.46%	6.70%	11.47%
Mean ROK small firms	4.79%	3.67%	4.79%	3.67%

Panel B: Parameter estimates

	$a^\star$	η	α	β	γ	Θ	δ_n	κ	ϑ	ν	Υ	F	f
Period 1	1.960	0.054	0.215	0.993	10.18	260.87	2.40%	0.008	62.5	6.27	1.774	0.527	0.034
	(0.150)	(0.007)	(0.005)	(0.001)	(1.59)	(123.21)	(0.03%)	(0.002)	(11.1)	(0.11)	(0.025)	(0.090)	(0.005)
Period 2	0.832	0.222	0.200	0.998	6.82	364.89	2.09%	0.045	122.9	6.24	1.944	0.403	0.015
	(0.058)	(0.016)	(0.008)	(0.000)	(0.64)	(76.98)	(0.03%)	(0.007)	(6.0)	(0.17)	(0.050)	(0.029)	(0.001)
Diff.	−1.128	0.168	−0.015	0.00	−3.36	104.02	−0.31%	0.036	60.4	−0.03	0.170	−0.125	−0.019

5.4 Table 3: Target moments and parameter estimates

Read: The model matches the rise in Q polarization, the jump in superstar markups, the rise in intangible capital, lower real rates, and slower output growth. Parameter estimates show rising small-firm entry cost, higher difficulty of breakthrough innovation, and rising taste for superstar goods.

Economic message. The model fits both macro trends and superstar-small-firm divergence.

5.5 Table 4: Model moments

Read: Panel B shows the model closely reproduces the increase in Q, markup, profitability, and intangible-capital polarization while keeping sales-growth and discount-rate differences small.

Economic message. This table validates the model's central empirical target.

Table 4
Model moments.

Panel A. Aggregate

	Data	Model		Data	Model
I. Total factor productivity			III. Consumption and output		
$\sigma[\Delta tfp]$	1.48%	1.52%	E$[\Delta y]$	1.70%	1.70%
AC1$[E[\Delta tfp]]$	0.944	0.928	$\sigma[\Delta c]$	0.98%	1.14%
			$\sigma[\Delta y]/\sigma[\Delta c]$	1.48	1.67
II. Investment			$\sigma[\Delta l]/\sigma[\Delta c]$	0.61	0.45
E$[I/K]$	9.26%	13.75%	AC1$[\Delta c]$	0.459	0.366
E$[S/Z]$	32.09%	30.77%	AC1$[\Delta y]$	0.345	0.049
$\sigma[I/K]$	1.07%	0.58%	AC1$[\Delta l]$	0.005	−0.064
$\sigma[S/Z]/\sigma[I/K]$	1.41	1.41			
AC1$[I/K]$	0.978	0.927			
AC1$[S/Z]$	0.964	0.925			

Panel B: Industry polarization

	Data			Model		
	1978–1998	1999–2019	Diff.	1978–1998	1999–2019	Diff.
ΔQ	0.28	1.39	1.11‡	0.28	1.36	1.08‡
ΔMarkup	9.84%	32.43%	22.58%‡	9.92%	32.54%	22.62%‡
ΔProfitability	1.90%	7.82%	5.92%‡	1.92%	7.80%	5.88%‡
ΔIntangible ratio	0.10	0.44	0.34‡	0.10	0.43	0.33‡
ΔSales growth	0.60%	0.06%	−0.54%	0.04%	0.05%	0.01%
ΔDiscount rate	1.73%	2.14%	0.42%	−0.21%	−0.23%	−0.02%

This table presents several model moments compared with data. Panel A focuses on key macroeconomic moments, obtained by averaging across the two subperiods: expected total factor productivity growth $E[\Delta tfp]$, calculated using the method of Bansal et al. (2007) by projecting one-period-ahead productivity growth on the risk-free rate and the log price-dividend ratio; the investment rate in physical capital (I/K) and intangible capital (S/Z); and the growth rates of aggregate consumption (Δc), output (Δy), and labor (Δl). Panel B reports key industry polarization variables, calculated as the difference between superstar and small firms in each subperiod. The 'Diff.' column shows the difference between the second and first subperiods, with statistical significance (Newey–West adjusted) denoted by *, †, and ‡ for the 10%-, 5%-, and 1%-level, respectively. Sales growth in the data is normalized to have the same volatility as output growth in the model. The reported model moments are computed for each subperiod using 200 samples of the same length as the data, with a burn-in period of 50 quarters. Growth rate moments are annualized percentages. $AC1$ denotes first-order autocorrelations. Details regarding the definition and construction of variables are provided in internet appendix C.

Table 5
Sources of Q polarization.

	Θ, ϑ	δ_n, κ	$\Theta, \vartheta, \delta_n, \kappa$	$\Theta, \vartheta, \delta_n, \kappa, \nu, Y$	All parameters
$\bar{Q} - \bar{\mathrm{Q}}$	0.01	0.34	0.38	0.86	1.08
$\Phi - \varphi$	3.17%	4.48%	16.88%	22.85%	22.62%
$\overline{SK} - \overline{sk}$	−2.68%	2.87%	3.62%	8.67%	32.53%
$\overline{FC} - \overline{fc}$	−0.01	0.01	−0.00	−0.00	0.05
$\bar{R} - \bar{r}$	0.24%	−0.02%	0.23%	0.15%	−0.03%
$\bar{G} - \bar{g}$	−0.05%	0.00%	−0.06%	−0.05%	0.01%
$\Delta_n - \delta_n$	−0.50%	1.24%	0.74%	0.74%	0.74%
$\bar{Z}/\bar{K} - \bar{z}/\bar{k}$	−0.04	0.01	−0.03	−0.03	0.34

This table investigates the contribution of various deep model parameters to the variables driving the polarization in Q, namely, the difference in price markup ($\Phi - \varphi$), sales-to-capital ratio ($\overline{SK} - \overline{sk}$), fixed cost of production ($\overline{FC} - \overline{fc}$), discount rates ($\bar{R} - \bar{r}$), growth rates ($\bar{G} - \bar{g}$), death rates ($\Delta_n - \delta_n$), and intangible capital share ($\bar{Z}/\bar{K} - \bar{z}/\bar{k}$). The contribution of each parameter set is determined by setting the parameter values to those of the 1999–2019 sample, while keeping all other parameters at their 1978–1998 values. All reported values are relative differences from the corresponding 1978–1999 sample moments, computed using simulated data from the model. Growth rates, death rates and returns are presented as annualized percentage rates. Details regarding the definition and construction of variables can be found in internet appendix C.

5.6 Table 5: Sources of Q polarization

Read: Small-firm entry barriers explain about 0.34 of Q polarization; superstar-transition barriers raise markups but explain little Q polarization. Adding taste parameters raises the Q contribution to 0.86, while all parameters generate 1.08.

Economic message. Markup polarization and Q polarization have different structural sources.

5.7 Figure 2: Impulse response functions entry costs

Read: The figure compares shocks to superstar-transition barriers and small-firm entry costs calibrated to reduce superstar markups. The former raises speculative innovation and growth; the latter reduces speculative innovation and growth.

Economic message. Pro-competitive policies are not equivalent: targeting large incumbents and encouraging small-firm entry can have opposite innovation and growth effects.

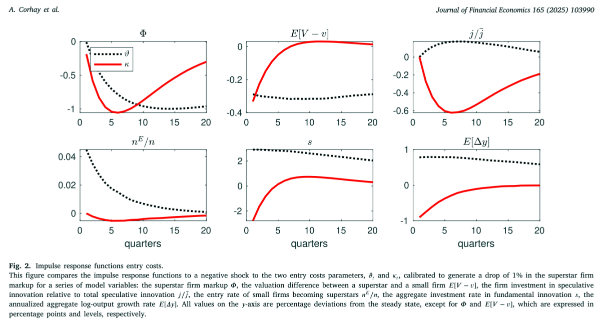

Figure 2. Impulse response functions entry costs.

6 Short summary

- Q polarization within industries rises sharply after the late 1990s.
- The rise is matched by polarization in markups, profitability, and intangibles, but not in growth or discount rates.
- The model features superstar oligopolists, small firms, endogenous transitions, speculative innovation, strategic R&D, and knowledge spillovers.
- Rising barriers to entry explain most markup polarization.
- Small-firm entry barriers, superstar demand shifts, and productive intangibles explain most Q polarization.
- Policy experiments show that different competition policies have different implications for innovation and growth.

7 Conclusion

7.1 Core conclusion

Corhay, Kung, and Schmid show that rising Tobin's Q is not only an aggregate valuation puzzle. It is also a within-industry inequality puzzle generated by superstar firms pulling away from their industry peers.

7.2 Mechanism

The mechanism is the interaction between entry barriers and innovation incentives. Higher barriers to creating small firms shrink the small-firm sector and increase superstar market shares. This raises the reward to becoming a superstar, stimulates speculative innovation, and strengthens knowledge spillovers. Higher barriers to becoming a superstar consolidate incumbent market power but discourage transition innovation and can lower growth.

7.3 Policy implications

Reducing superstar barriers can reduce markup inequality and stimulate growth. Reducing small-firm entry barriers can reduce Q and markup inequality but may weaken speculative innovation and trend growth. Competition policy therefore needs to account for the life-cycle stage at which barriers operate.

7.4 Limitations

The results are structural and depend on the model assumptions about demand, oligopoly, entry, and spillovers. The policy experiments are reduced-form shocks to model parameters rather than direct institutional reforms.

7.5 Takeaway

Q polarization is informative because it embeds expectations about rents, intangibles, innovation, and growth. It helps distinguish why markups rise from why valuations diverge.